



Graphic Era
HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Act Number 12 of 2011)
University under section 2(f) of UGC Act, 1956

22-61514

End Term Back Paper Examination January 2025

Name of the course: B.Tech

Name of the paper: Engineering Mathematics - II

Time: 3 hours

Semester: I

Paper Code: TMA -201

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b and c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each sub part carries 10 marks.

Q.1

(10×2 = 20 Marks) CO: 1

- a. Solve of $(1+x^2)dy - xy dx = 0$.
- b. Test for exactness and solve the differential equation, $(y^2 - x^2)dx + 2xydy = 0$.
- c. Solve by the method of variation of parameters: $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$.

Q.2

(10×2 = 20 Marks) CO: 2

- a. Find the Laplace transform of $\frac{1-\cos t}{t}$.
- b. Using convolution theorem prove that $L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\} = \frac{a \sin at - b \sin bt}{a^2 - b^2}$.
- c. Find the Laplace transform of the periodic function $f(t) = e^t$ for $0 < t < 2\pi$.

Q.3

(10×2 = 20 Marks) CO: 3

- a. Expand the function $f(x) = x^2$ as a Fourier series in the interval $-\pi \leq x \leq \pi$.
- b. Find the Fourier series expansion of the periodic function of period 2π .
 $f(x) = e^x, \quad 0 < x < 2\pi$
- c. Obtain the Fourier series expansion of $f(x) = \begin{cases} x & 0 < x < \pi \\ -x, & -\pi < x < 0 \end{cases}$



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Q.4

(10×2=20 Marks) CO: 4

- a. Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = \cos(2x + y)$
- b. Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$; $u(x, 0) = 6e^{-3x}$
- c. Find the temperature in a bar of length 2 whose ends kept at zero and lateral surface insulated if initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.

Q.5

(10×2=20 Marks) CO: 5

- a. Prove that: $\int_{-1}^1 P_m(x) P_n(x) dx = 0$, if $m \neq n$
- b. Prove that: $x J_n'(x) = n J_n(x) - x J_{n-1}(x)$
- c. Prove that: $(2n+1)x P_n = (n+1) P_{n+1} + n P_{n-1}$